



PEOPLE

13 Teenagers Who Are Changing The World

Technology has always been the bastion of the young, and to celebrate our teen birthday, here are some impressive teens

Samir Alam

1 Nathan Han

The most recent whizkid of our generation, Nathan Han, a student of the Boston Latin School, proved his intellectual mettle by placing first at Intel's International Science and Engineering Fair in May 2014. He used publicly available databases on gene research to examine the characteristics of the various mutations of the BRCA1 tumor suppressor gene to create his software. Using the data this software was "taught" how to differentiate between different gene mutations in order to spot the ones that cause cancer. His inventive approach to mixing biology, software and statistical analysis has presented scientists with a whole new way of fighting cancer. Nathan looks just like any 9th grader, but has already made his mark on the world. He's shown that with imagination, support and determination even the most amazing discoveries can be achieved from just a teenager's laptop.



2 Sara Volz

Sara Volz was just 17 years old when she won the Intel Science Talent Search in 2013. Humorously known as an algae enthusiast, she cultivated an algae experiment under her bed. At the time she was a student of Cheyenne Mountain High School in Colorado Springs where she was just about to graduate when her project won her the Talent Search and a \$100,000 prize that she used to go to the Massachusetts Institute of Technology. The young researcher has gone on to advocate for the cause of algae based biofuels as a replacement for fossil fuels by speaking at major event including giving a TED Talk. Her biggest



impact is expected in her work towards developing ways to make algae-based biofuel more cost effective and efficiently produced by using "artificial selection" techniques to grow algae. Her commitment to the field has already shown potential to radically change the world and she's only getting started.

3 Ari Dyckovsky

Ari Dyckovsky began solving complex mathematical problems at age 4 and by 18 was dabbling in the field of quantum physics at the Loudoun Academy of Science. At 18 he published a paper in the "Physics Review A" that shattered the ceiling on research being done towards the creation of quantum computers and quantum entanglement. He intends to create an instantaneous data transfer technology which would solve the problems of speed and security. He dropped out of Stanford to start a company called Arktos. He's taking on giants such as Microsoft by creating cutting edge data visualisation programs. Dyckovsky's innovations have already altered the expanse of science and technology.



4 Angela Zhang

As a 15 year old student in the 9th grade Angela Zhang began taking an interest in bioengineering – by the time she was in 11th grade she had won USD 100,000 at the National Siemens Science Contest for her science project on devel-



oping a cure for cancer. With access to research laboratories in Stanford, she spent nearly a thousand hours of research before successfully creating a nanoparticle that is capable of attacking cancer cells and also providing diagnostic information. She even figured out how to combine cancer medication within a polymer attached to these nanoparticles. Early testing has proven successful in rats, and in a few years she could be credited with curing cancer!

5 Taylor Wilson

As a 16 year old, Taylor Wilson won a host of prizes at the 2010 Intel International Science and Engineering Fair with his project titled "Fission Vision: The Detection of Prompt and Delayed Induced Fission Gamma, Radiation, and the Application to the Detection of Proliferated Nuclear Materials" (that's a radiation detector to us common folk). The US Department of Homeland Security and the US Department of Energy offered him funding to build a low-cost Cherenkov radiation detector for border security and anti-terrorist threats. Wilson then went on to become a bona fide nuclear physicist at just 18! He's working on a low-cost waste fueled nuclear reactor his main priority.

